CSE370 : Database Systems

Assignment 01 | Spring 2026

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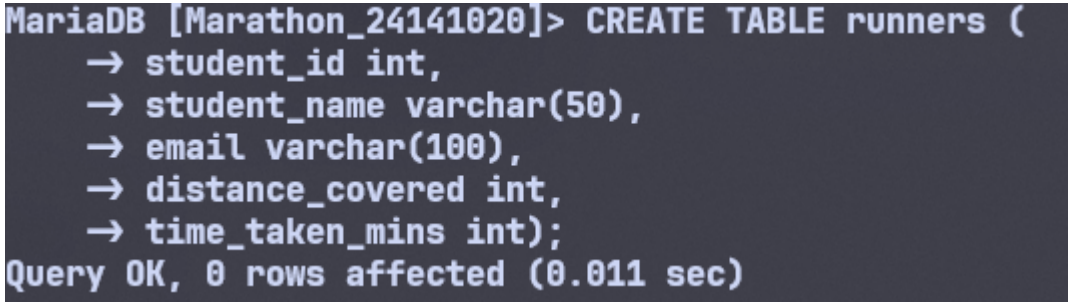
No 1 Query (as Plain Text)	CREATE TABLE runners (-> student_id int, -> student_name varchar(50), -> email varchar(100), -> distance_covered int, -> time_taken_mins int);
No 1 SS (of Query & Output in Shell)	

```
MariaDB [Marathon_24141020]> SHOW DATABASES;
+--------------------+
| Database           |
+--------------------+
| 24141020           |
| information_schema |
| marathon_24141020  |
| mysql              |
| performance_schema |
| phpmyadmin         |
| test               |
+--------------------+
7 rows in set (0.000 sec)
```

```
MariaDB [Marathon_24141020]> show tables;
+-----------------------------+
| Tables_in_marathon_24141020 |
+-----------------------------+
| runners                     |
+-----------------------------+
1 row in set (0.000 sec)

MariaDB [Marathon_24141020]> describe runners;
+------------------+--------------+------+-----+---------+-------+
| Field            | Type         | Null | Key | Default | Extra |
+------------------+--------------+------+-----+---------+-------+
| student_id       | int(11)      | YES  |     | NULL    |       |
| student_name     | varchar(50)  | YES  |     | NULL    |       |
| email            | varchar(100) | YES  |     | NULL    |       |
| distance_covered | int(11)      | YES  |     | NULL    |       |
| time_taken_mins  | int(11)      | YES  |     | NULL    |       |
+------------------+--------------+------+-----+---------+-------+
5 rows in set (0.011 sec)
```

No 2 Query (as Plain Text)

```
insert into runners (student_id, student_name, email, distance_covered, time_taken_mins) VALUES
   -> (101, 'Mikasa Ackerman', 'mikasa.a@g.bracu.ac.bd', 5, 35),
   -> (102, 'Edward Elric', 'e.elric@iub.ac.bd', 10, 59),
   -> (103, 'Riza Hawkeye', 'riza.h@ewu.ac.bd', 5, 38),
   -> (104, 'Bruce Wayne', 'b.wayne@g.bracu.ac.bd', 10, 63),
   -> (105, 'Peter Parker', 'p.parker@iub.ac.bd', 5, 33),
```

	-> (106, 'Alphonse Elric', 'a.elric@iub.ac.bd', 10, 55), -> (107, 'Son Goku', 'son.goku@g.bracu.ac.bd', 5, 46), -> (108, 'Gon Freecss', 'gon.f@g.bracu.ac.bd', 10, 68), -> (109, 'Tanjiro Kamado', 'tanjiro.k@g.bracu.ac.bd', 5, 29), -> (110, 'Roy Mustang', 'roy.m@ewu.ac.bd', 10, 83);
No 2 SS (of Query & Output in Shell)	MariaDB [Marathon_24141020]> insert into runners (student_id, student_name, email, distance_covered, time_taken_mins) VALUES -> (101, 'Mikasa Ackerman', 'mikasa.a@g.bracu.ac.bd', 5, 35), -> (102, 'Edward Elric', 'e.elric@iub.ac.bd', 10, 59), -> (103, 'Riza Hawkeye', 'riza.h@ewu.ac.bd', 5, 38), -> (104, 'Bruce Wayne', 'b.wayne@g.bracu.ac.bd', 10, 63), -> (105, 'Peter Parker', 'p.parker@iub.ac.bd', 5, 33), -> (106, 'Alphonse Elric', 'a.elric@iub.ac.bd', 10, 55), -> (107, 'Son Goku', 'son.goku@g.bracu.ac.bd', 5, 46), -> (108, 'Gon Freecss', 'gon.f@g.bracu.ac.bd', 10, 68), -> (109, 'Tanjiro Kamado', 'tanjiro.k@g.bracu.ac.bd', 5, 29), -> (110, 'Roy Mustang', 'roy.m@ewu.ac.bd', 10, 83); Query OK, 10 rows affected (0.009 sec) Records: 10 Duplicates: 0 Warnings: 0

```
MariaDB [Marathon_24141020]> describe runners;
+------------------+--------------+------+-----+---------+-------+
| Field            | Type         | Null | Key | Default | Extra |
+------------------+--------------+------+-----+---------+-------+
| student_id       | int(11)      | YES  |     | NULL    |       |
| student_name     | varchar(50)  | YES  |     | NULL    |       |
| email            | varchar(100) | YES  |     | NULL    |       |
| distance_covered | int(11)      | YES  |     | NULL    |       |
| time_taken_mins  | int(11)      | YES  |     | NULL    |       |
+------------------+--------------+------+-----+---------+-------+
5 rows in set (0.008 sec)

MariaDB [Marathon_24141020]> select * from runners;
+------------+-----------------+-------------------------+------------------+-----------------+
| student_id | student_name    | email                   | distance_covered | time_taken_mins |
+------------+-----------------+-------------------------+------------------+-----------------+
|        101 | Mikasa Ackerman | mikasa.a@g.bracu.ac.bd  |                5 |              35 |
|        102 | Edward Elric    | e.elric@iub.ac.bd       |               10 |              59 |
|        103 | Riza Hawkeye    | riza.h@ewu.ac.bd        |                5 |              38 |
|        104 | Bruce Wayne     | b.wayne@g.bracu.ac.bd   |               10 |              63 |
|        105 | Peter Parker    | p.parker@iub.ac.bd      |                5 |              33 |
|        106 | Alphonse Elric  | a.elric@iub.ac.bd       |               10 |              55 |
|        107 | Son Goku        | son.goku@g.bracu.ac.bd  |                5 |              46 |
|        108 | Gon Freecss     | gon.f@g.bracu.ac.bd     |               10 |              68 |
|        109 | Tanjiro Kamado  | tanjiro.k@g.bracu.ac.bd |                5 |              29 |
|        110 | Roy Mustang     | roy.m@ewu.ac.bd         |               10 |              83 |
+------------+-----------------+-------------------------+------------------+-----------------+
10 rows in set (0.000 sec)
```

No 3 Query (as Plain Text)	select * from runners where distance_covered > 6;

No 3 SS
(of Query & Output in Shell)

```
MariaDB [Marathon_24141020]> select * from runners where distance_covered > 6;
+------------+---------------+----------------------+------------------+-----------------+
| student_id | student_name  | email                | distance_covered | time_taken_mins |
+------------+---------------+----------------------+------------------+-----------------+
|        102 | Edward Elric  | e.elric@iub.ac.bd    |               10 |              59 |
|        104 | Bruce Wayne   | b.wayne@g.bracu.ac.bd |               10 |              63 |
|        106 | Alphonse Elric | a.elric@iub.ac.bd    |               10 |              55 |
|        108 | Gon Freecss   | gon.f@g.bracu.ac.bd  |               10 |              68 |
|        110 | Roy Mustang   | roy.m@ewu.ac.bd      |               10 |              83 |
+------------+---------------+----------------------+------------------+-----------------+
5 rows in set (0.000 sec)
```

No 4 Query
(as Plain Text)

select upper(student_name) as student_name from runners order by time_taken_mins desc;

No 4 SS
(of Query & Output in Shell)

```
MariaDB [Marathon_24141020]> select upper(student_name) as student_name from runners order by ti
me_taken_mins desc;
+----------------+
| student_name   |
+----------------+
| ROY MUSTANG    |
| GON FREECSS    |
| BRUCE WAYNE    |
| EDWARD ELRIC   |
| ALPHONSE ELRIC |
| SON GOKU       |
| RIZA HAWKEYE   |
| MIKASA ACKERMAN |
| PETER PARKER   |
| TANJIRO KAMADO |
+----------------+
10 rows in set (0.000 sec)
```

No 5 Query (as Plain Text)	alter table runners change distance_covered total_distance decimal(3,1);

No 5 SS (of Query & Output in Shell)	`MariaDB [Marathon_24141020]> alter table runners change distance_covered total_distance decimal(3,1);`
	`Query OK, 10 rows affected (0.045 sec)`
	`Records: 10  Duplicates: 0  Warnings: 0`
	
	`MariaDB [Marathon_24141020]> describe runners;`
	`+-----+---------+---------+---------+---------+---------+`
	`| Field | Type | Null | Key | Default | Extra |`
	`+-----+---------+---------+---------+---------+---------+`
	`| student_id | int(11) | YES | | NULL | |`
	`| student_name | varchar(50) | YES | | NULL | |`
	`| email | varchar(100) | YES | | NULL | |`
	`| total_distance | decimal(3,1) | YES | | NULL | |`
	`| time_taken_mins | int(11) | YES | | NULL | |`
	`+-----+---------+---------+---------+---------+---------+`
	`5 rows in set (0.009 sec)`

No 6 Query (as Plain Text)	alter table runners add kmh decimal(3,1);

No 6 SS (of Query & Output in Shell)	`MariaDB [Marathon_24141020]> alter table runners add kmh decimal(3,1);`
	`Query OK, 0 rows affected (0.006 sec)`
	`Records: 0  Duplicates: 0  Warnings: 0`

```
MariaDB [Marathon_24141020]> describe runners;
+-----------------+--------------+------+-----+---------+-------+
| Field           | Type         | Null | Key | Default | Extra |
+-----------------+--------------+------+-----+---------+-------+
| student_id      | int(11)      | YES  |     | NULL    |       |
| student_name    | varchar(50)  | YES  |     | NULL    |       |
| email           | varchar(100) | YES  |     | NULL    |       |
| total_distance  | decimal(3,1) | YES  |     | NULL    |       |
| time_taken_mins | int(11)      | YES  |     | NULL    |       |
| kmh             | decimal(3,1) | YES  |     | NULL    |       |
+-----------------+--------------+------+-----+---------+-------+
6 rows in set (0.008 sec)
```

No 7 Query (as Plain Text)	update runners set kmh = (total_distance * 60) / time_taken_mins;

No 7 SS (of Query & Output in Shell)	`MariaDB [Marathon_24141020]> update runners set kmh = (total_distance * 60) / time_taken_mins;`
	`Query OK, 10 rows affected, 10 warnings (0.003 sec)`
	`Rows matched: 10  Changed: 10  Warnings: 10`

```
MariaDB [Marathon_24141020]> select * from runners;
+------------+----------------+-------------------------+----------------+-----------------+------+
| student_id | student_name   | email                   | total_distance | time_taken_mins | kmh  |
+------------+----------------+-------------------------+----------------+-----------------+------+
|        101 | Mikasa Ackerman | mikasa.a@g.bracu.ac.bd  |            5.0 |              35 |  8.6 |
|        102 | Edward Elric   | e.elric@iub.ac.bd       |           10.0 |              59 | 10.2 |
|        103 | Riza Hawkeye   | riza.h@ewu.ac.bd        |            5.0 |              38 |  7.9 |
|        104 | Bruce Wayne    | b.wayne@g.bracu.ac.bd   |           10.0 |              63 |  9.5 |
|        105 | Peter Parker   | p.parker@iub.ac.bd      |            5.0 |              33 |  9.1 |
|        106 | Alphonse Elric | a.elric@iub.ac.bd       |           10.0 |              55 | 10.9 |
|        107 | Son Goku       | son.goku@g.bracu.ac.bd  |            5.0 |              46 |  6.5 |
|        108 | Gon Freecss    | gon.f@g.bracu.ac.bd     |           10.0 |              68 |  8.8 |
|        109 | Tanjiro Kamado | tanjiro.k@g.bracu.ac.bd |            5.0 |              29 | 10.3 |
|        110 | Roy Mustang    | roy.m@ewu.ac.bd         |           10.0 |              83 |  7.2 |
+------------+----------------+-------------------------+----------------+-----------------+------+
10 rows in set (0.000 sec)
```

No 8 Query (as Plain Text)	select student_id, student_name, kmh, (total_distance / 10) * (time_taken_mins / 60) * 100 as points from runners;

No 8 SS (of Query & Output in Shell)	

```
MariaDB [Marathon_24141020]> select student_id, student_name, kmh, (total_distance / 10) * (time_taken_m
ins / 60) * 100 as points from runners;
+------------+-----------------+------+---------------+
| student_id | student_name    | kmh  | points        |
+------------+-----------------+------+---------------+
|        101 | Mikasa Ackerman |  8.6 |  29.166666650 |
|        102 | Edward Elric    | 10.2 |  98.333333300 |
|        103 | Riza Hawkeye    |  7.9 |  31.666666650 |
|        104 | Bruce Wayne     |  9.5 | 105.000000000 |
|        105 | Peter Parker    |  9.1 |  27.500000000 |
|        106 | Alphonse Elric  | 10.9 |  91.666666600 |
|        107 | Son Goku        |  6.5 |  38.333333300 |
|        108 | Gon Freecss     |  8.8 | 113.333333300 |
|        109 | Tanjiro Kamado  | 10.3 |  24.166666650 |
|        110 | Roy Mustang     |  7.2 | 138.333333300 |
+------------+-----------------+------+---------------+
10 rows in set (0.000 sec)
```

No 9 Query (as Plain Text)	select student_name from runners where total_distance = 10 and time_taken_mins between 40 and 60;
No 9 SS (of Query & Output in Shell)	

```
MariaDB [Marathon_24141020]> select student_name from runners where total_distance = 10 and time_taken_m
ins between 40 and 60;
+----------------+
| student_name   |
+----------------+
| Edward Elric   |
| Alphonse Elric |
+----------------+
2 rows in set (0.000 sec)
```

No 10 Query	select student_name from runners where email like '%g.bracu.ac.bd';

(as Plain Text)	
No 10 SS (of Query & Output in Shell)	

```
MariaDB [Marathon_24141020]> select student_name from runners where email like '%g.bracu.ac.bd';
+-----------------+
| student_name    |
+-----------------+
| Mikasa Ackerman |
| Bruce Wayne     |
| Son Goku        |
| Gon Freecss     |
| Tanjiro Kamado  |
+-----------------+
5 rows in set (0.000 sec)
```